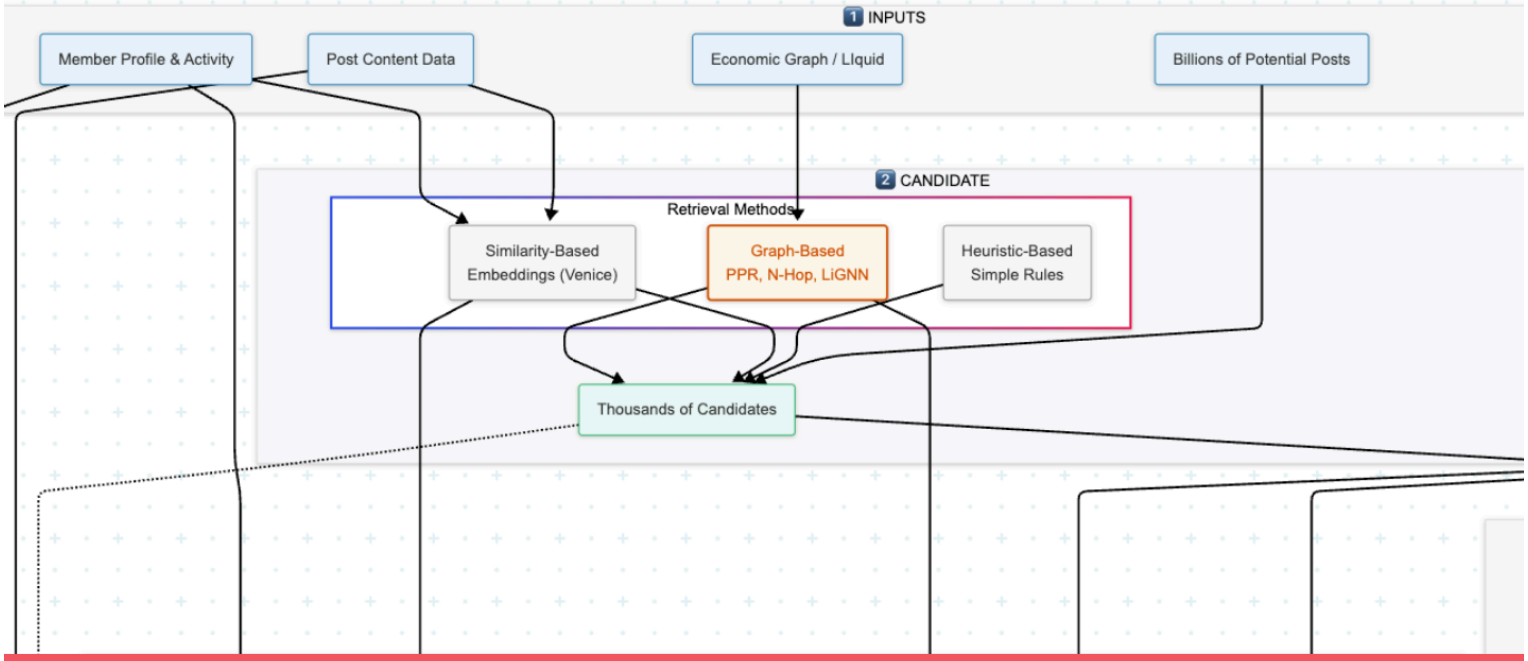
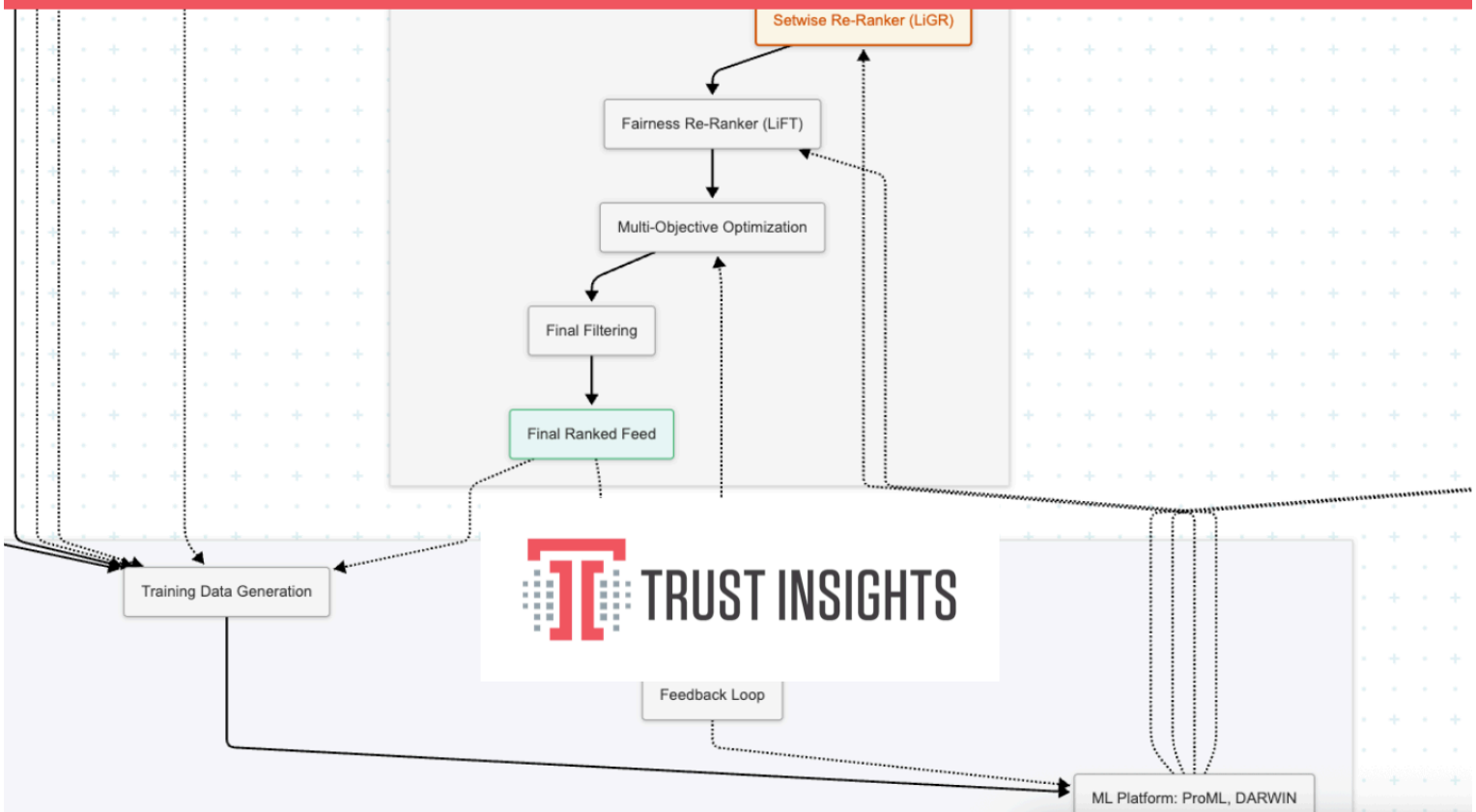


The Unofficial LinkedIn Algorithm Guide for Marketers

Not endorsed, reviewed, or approved by LinkedIn!

Mid 2025 Edition



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Introduction

If you've spent more than 5 minutes on LinkedIn, you've doubtlessly seen one or more "gurus" making claims that they've "hacked the LinkedIn algorithm" and if you only comment on their time-limited post, they'll sell you the secret to boosting your performance on LinkedIn.

That's not how LinkedIn works.

The reality is that LinkedIn's newsfeed isn't an algorithm. It's a series of **FOURTEEN** very complex, very technical services woven together in a global infrastructure that makes some fairly amazing decisions at lightning speed. ***What that means for you and me is that there's no "hack" for more than a dozen different AI systems interoperating with each other.***

What that also means is that if we understand what those systems are, how they work, and what they take into account, we can follow some basic best practices to work with them, not against them. This isn't hacking anything. This is aligning our efforts with the way the systems are built.

So how would we do this? By listening to what LinkedIn has had to say. In this totally unofficial guide to LinkedIn's algorithm, we've used generative AI to boil down dozens of articles, papers, interviews, and conference submissions from Linkedin employees to build a comprehensive explanation of what happens to our content as LinkedIn processes it. Each step of the process details what we can do to best work with the algorithms (yes, plural - there are many) to be featured.

This guide is not endorsed or approved by LinkedIn; no LinkedIn employees were consulted outside of using public data published by those employees.

After the walkthrough, you'll find THREE toolkits you can copy and paste into your favorite generative AI tool:

The LinkedIn Profile Checklist

Use this toolkit and your LinkedIn profile to get suggestions for improving your LinkedIn profile according to LinkedIn's system.

The LinkedIn Content Pre-Launch Checklist

Use this toolkit to validate content before you post it on LinkedIn, especially if it's pillar content you want high visibility on.

The LinkedIn Engagement Checklist

Use this toolkit to guide your activities every day and week, helping you build the foundation for more visibility on everything you do.

At the end of the guide is an explanation of the technical details and the sources we used. This edition is significantly updated because of two MAJOR new papers LinkedIn published for the ACD KDD conferences in 2024 and 2025.

Trademarks, copyrights, and other intellectual property remain the property of their stakeholders, used under fair use journalistic laws.

Because we made this with generative AI using Google's Gemini 2.5 Advanced 0504 model, it is entirely possible there are hallucinations (statistically correct but factually wrong information) within this guide, despite our best efforts to check it. Use it as a guide, not a rigid set of rules.

This guide was last updated May 20, 2025.

How LinkedIn's Algorithms **Actually** Work

I. Inputs Gathering & Preprocessing

Step 1: Your Content's Initial Ingestion & Analysis

- **What happens**: When you post content (text, image, video, article, link), LinkedIn's systems first take it in. Sophisticated AI (including technologies like DeText and potentially newer Large Language Models like EON/360Brew for some feature generation) then analyze this content.
    - They break down your text into understandable components, identify key topics, entities (like companies or skills mentioned), and even try to understand the sentiment.
    - Images and videos are similarly analyzed for objects, scenes, and overall themes.
    - All this information, including your raw content and these AI-derived insights (like embeddings – a kind of digital fingerprint), is then stored in fast-access databases (like Venice).
- **So what?**: This is the first impression your content makes on the *system*. If the AI can't easily understand what your post is about or who it's relevant to, it's harder for it to be matched effectively later on. Clear, well-defined content gets a better "digital fingerprint."
- **Now what?**:
    - **Be Clear and Specific:** Ensure your text is well-written and clearly communicates the core message. Use unambiguous language.
    - **Relevant Keywords & Entities:** Naturally incorporate relevant keywords, industry terms, and mention relevant companies or skills if appropriate. This helps the AI categorize and understand your content's context. Don't keyword stuff; make it organic.
    - **High-Quality Media:** Use clear, high-resolution images and well-produced videos. Ensure any text within images is legible. Provide descriptive alt-text for images.

Step 2: Understanding the Member (Your Audience)

- **What happens**: LinkedIn continuously processes information about its members. This includes:

    - **Profile Data:** Information members provide like their job title, industry, skills, company, education, and connections. This data is stored and structured (likely in systems like Espresso and reflected in the Economic Graph/LIquid).
    - **Activity History:** What members view, click, like, comment on, share, search for, and how long they spend on different types of content. This activity is captured in real-time (via Kafka) and processed both quickly for recent actions (in systems like Pinot) and over longer periods for deeper patterns (using Spark on HDFS, with results stored in Venice).
    - **Embeddings:** AI creates "digital fingerprints" (embeddings) for members based on their profile and activity, representing their professional identity and interests.
- **So what?**: Your content needs to resonate with specific member profiles and their demonstrated interests. LinkedIn is trying to match your content to members whose "digital fingerprints" align with what your content offers.
- **Now what?**:
    - **Know Your Target Audience Deeply:** Understand their typical job titles, industries, skills, and the kind of content they usually engage with on LinkedIn.
    - **Align Content with Audience Interests:** Create content that directly addresses the needs, challenges, or interests of your target audience's professional lives.
    - **Consider Your Own Profile:** Ensure your own profile accurately reflects your expertise and the audience you want to reach, as your identity is also part of the matching equation.

Step 3: Leveraging the LinkedIn Economic Graph

- **What happens**: LinkedIn maintains a massive "Economic Graph" (powered by LIquid) that maps relationships between members, companies, skills, schools, etc. This graph is constantly updated.
- **So what?**: The system uses this graph to understand connections and context. If your content or your profile has strong, relevant connections within this graph (e.g., you're connected to people in the industry you're posting about, or your post mentions a company relevant to a target audience's network), it signals relevance.
- **Now what?**:
    - **Build a Relevant Network:** Connect with professionals in your target industry and individuals who engage with similar content.

- **Tag Relevant Entities:** When appropriate, tag relevant companies or individuals (influencers, collaborators) in your posts to explicitly link your content into the graph.
- **Use Relevant Hashtags:** Hashtags help categorize your content and connect it to broader conversations and interest groups within the graph.

II. L0: Candidate Generation (Getting on the Initial Longlist)

Step 4: Graph-Based Candidate Retrieval

- **What happens**: LinkedIn uses its understanding of the professional network (the Economic Graph, queried via LIquid) to find initial potential posts for a member's feed.
  - It looks at posts from your direct connections, connections of your connections (2nd-degree), and even further out (3rd-degree).
  - It also uses "Personalized PageRank" to find posts from influential or highly relevant people within your extended network.
  - Sophisticated Graph Neural Networks (GNNs, like in LiGNN) create advanced "graph embeddings" that capture complex relationships. These embeddings are used to find posts from people or about topics that are structurally similar or related to the member's position in the graph, even if not directly connected. This happens through "Embedding Based Retrieval" (EBR).
- **So what?**: Being connected, even indirectly, to your target audience matters significantly. The more relevant your network and the network of people engaging with your content, the higher the chance your post gets picked up in this initial sweep. GNNs can find your content if its *topic or creator profile* is similar to what the member usually interacts with, based on graph structure.
- **Now what?**:
  - **Encourage Engagement from Your Network:** Likes, comments, and shares from your direct connections significantly boost your post's visibility to *their* connections, which are often your 2nd or 3rd-degree targets.
  - **Content Creator's Network Matters:** The system looks at *your* network quality and the network of people interacting with your post. If your engagers are relevant to the target viewer, it helps.
  - **Be Discoverable via Graph Similarity:** Create content and maintain a profile that aligns with the "graph neighborhood" of your target audience. If your skills, industry, and connections overlap with theirs, GNN-based retrieval is more likely to find you.

Step 5: Similarity-Based Candidate Retrieval (EBR)

- **What happens**: Beyond just network connections, LinkedIn creates "digital fingerprints" (embeddings) for members (based on their profile and activity) and for every piece of content.
    - When a member opens their feed, the system takes their "member embedding" and searches a massive database (using systems like Hosted Search/Zelda and Venice) for posts with similar "post embeddings."
    - "Similar" means the content topics, style, or implied interests align.
- **So what?**: Your content needs a clear "digital fingerprint" that matches the "fingerprint" of your target audience's interests. If your post's embedding is close to a member's interest embedding, it's more likely to be pulled into their potential feed.
- **Now what?**:
    - **Thematic Consistency:** Create content that is thematically consistent with the topics your target audience is interested in.
    - **Mirror Successful Content (in theme, not verbatim):** Observe what kind of content (topics, formats) your target audience engages with and consider creating valuable content along similar themes.
    - **Clarity of Subject Matter:** The clearer your post's subject, the more distinct and matchable its embedding will be. Avoid overly broad or muddled topics in a single post if you want to be found via specific interest similarities.

Step 6: Heuristic-Based Candidate Retrieval

- **What happens**: Simpler, rule-based methods also pull in content. For example:
    - Posts from people a member has recently interacted with.
    - Very recent posts from their direct network that are gaining quick traction.
    - Content related to trending topics the member has shown interest in.
    - These rules are often executed against fast data stores like Pinot or Venice.
- **So what?**: Recent interactions and timeliness can give your content a temporary boost or ensure it's seen by those you've recently engaged with.
- **Now what?**:
    - **Engage Authentically:** Actively and thoughtfully engage with others in your target audience. This makes it more likely your own content will be surfaced to them via these heuristics.
    - **Timeliness for Trending Topics:** If your content relates to a currently trending topic, posting it in a timely manner can leverage these heuristic retrievals.

    - **Encourage Quick Engagement:** Initial engagement on a fresh post can trigger some of these heuristics.

III. L1: Light Ranking (Making the Shorter List)

Step 7: Initial Scoring & Calibration

- **What happens**: The thousands of posts gathered in L0 are now scored by a faster, "lighter" AI model (like XGBoost or a simple neural network, likely run via Quasar/ProML). This model uses a more limited set of features (basic post/author info, L0 source scores) to give each post a preliminary relevance score. It also helps to "calibrate" or normalize scores from the very different L0 sources.
- **So what?**: Your post needs to pass this first, broader relevance check. Even if it was picked up by one L0 source, if its basic features don't align well with the member's general profile at this stage, it might get filtered out.
- **Now what?**:
    - **Fundamental Relevance:** Ensure your post has fundamentally relevant features for your target audience (e.g., appropriate language, clear author identity, basic topical alignment).
    - **Strong L0 Signals:** The stronger your signals were in the L0 candidate generation (e.g., more common connections, higher embedding similarity), the better your L0 source score, which is a feature here.

IV. L2: Rich Ranking / SPR (The Main Relevance Engine)

Step 8: Deep Feature Gathering

- **What happens**: For the hundreds of posts that made it through L1, the system now gathers a much richer set of features. This includes:
    - **Member's Recent Interaction Sequence:** A detailed history of what the specific member has recently viewed, clicked, liked, commented on, etc. (from Pinot/Venice).
    - **Rich Candidate Post Features:** Deep "digital fingerprints" (embeddings) of your post's content (text, image, video), detailed metadata, and features of you, the author (from Venice).
    - **Rich Member Features:** Detailed "digital fingerprints" of the viewing member's profile and long-term interests (from Venice/Pensieve).

    - **Interaction Features:** How well *your specific post* matches *this specific member's* profile and interests, often calculated using embeddings (e.g., member-author affinity, member-post topic similarity).
    - **Real-Time Context:** What the member is doing *right now*, their device, etc.
- **So what?**: This is where deep personalization happens. The system is looking for a strong, multi-faceted match between your content, you as a creator, and the specific member viewing the feed, based on their very recent and historical behavior.
- **Now what?**:
    - **High-Quality, Engaging Content:** This is where the *quality* of your content truly shines. Content that elicits strong engagement signals (historically and potentially) will have features that resonate here.
    - **Author Credibility & Relevance:** Your own profile's relevance to the content and the viewing member is important. If you are a recognized expert or a relevant connection for that topic and member, it helps.
    - **Contextual Relevance:** Your post should ideally be relevant to what the member might be thinking about or doing *now* (if discoverable through real-time signals).

Step 9: Core Ranking Model Prediction (Transformer Power)

- **What happens**: These rich features are fed into a very powerful AI model, increasingly a **Transformer-based sequential model** (like in LiGR, running on ProML/Model Cloud).
    - This model "reads" the member's sequence of past interactions to understand their current and evolving interests.
    - It then predicts the probability of that specific member taking various actions on *your specific post* – actions like likes, comments, shares, clicks, reading for a long time, and skipping it.This is a "Multi-Task Learning" setup.
- **So what?**: Your content is being evaluated on its likelihood to drive *multiple types* of positive engagement (and avoid negative ones like a quick "skip") for *that specific member* given their *recent activity trail*. Content that aligns with the *trajectory* of a member's interests, as understood by the Transformer, will score higher.
- **Now what?**:
    - **Create "Action-Worthy" Content:** Think about what makes someone not just view, but *act* on a post (like, comment, share). Is it insightful, thought-provoking, helpful, conversation-starting?

- **Understand Engagement Patterns:** Observe what kind of content leads to deeper engagement (comments, shares) versus passive consumption (views, long dwells) in your niche and for your target audience.
- **Build a Narrative/Sequence:** If you post regularly, consider how your content builds a narrative or follows a logical sequence of interests that a Transformer model might pick up on from member engagement over time.
- **Optimize for Dwell Time (Implicitly):** Valuable, engaging content naturally leads to longer dwell times and lower skip rates, which are positive signals. Focus on substance.

V. Re-Ranking & Finalization

Step 10: Setwise Re-Ranking (Context is King)

- **What happens**: The top-scoring posts from L2 (e.g., the top 10-20) are now looked at *as a group* by another AI model (the Setwise Re-Ranker in LiGR). This model uses "setwise attention" to see how posts relate to *each other* in the potential feed.
  - It might boost a post that adds unique value or a different perspective to the set.
  - It might down-rank a post that's too similar to another, higher-ranked post already in the set.
- **So what?**: Even if your post scores highly on its own, its chances can be affected by what other posts are also ranking high for that member at that moment. Uniqueness and complementary value matter.
- **Now what?**:
  - **Offer a Unique Angle:** Try to provide a perspective or information that might be distinct from other common content on the same topic.
  - **Consider the "Feed Mix":** While you can't control other posts, be aware that if many similar posts are trending, having a slightly different take or format could help your content stand out in this stage.
  - **High Individual Score Still Key:** This stage refines the top; a strong individual score from L2 is necessary to even be considered here.

Step 11: Fairness Re-Ranking

- **What happens**: LinkedIn applies algorithms (potentially from its LiFT toolkit) to ensure that the distribution of content isn't unfairly biased towards or against certain groups of members or creators, based on protected attributes. Scores might be adjusted to promote more equitable exposure.

- **So what?**: This is largely an internal LinkedIn process to ensure platform health and fairness.
- **Now what?**:
    - **Focus on Broad Appeal (within your niche):** Creating high-quality, broadly valuable content for your intended professional audience is the best approach. Avoid content that could inadvertently perpetuate harmful stereotypes or biases.

Step 12: Multi-Objective Optimization (MOO) - The Balancing Act

- **What happens**: The various probability scores (P(Like), P(Click), etc.) for each post are now combined into a single final score. This is done using a weighted formula where LinkedIn assigns different levels of importance to different actions (e.g., a comment might be weighted higher than a like). These weights are carefully tuned (sometimes using automated Bayesian Optimization).
- **So what?**: LinkedIn is not just optimizing for one thing (like clicks). It's balancing multiple types of engagement to create a healthy ecosystem. Content that drives "deeper" or more "valuable" interactions (as defined by LinkedIn's current objectives) might get a higher final score even if its raw click probability isn't the absolute highest.
- **Now what?**:
    - **Aim for Meaningful Interactions:** Content that sparks discussions (comments) or is deemed valuable enough to reshare often aligns with LinkedIn's broader objectives for a professional community.
    - **Understand What LinkedIn Values:** Pay attention to the types of interactions LinkedIn highlights or encourages. This can give clues about what signals are weighted more heavily in the MOO. (e.g., emphasis on "knowledge sharing" suggests comments and insightful shares are valued).

Step 13: Final Filtering & Business Rules

- **What happens**: Before the feed is shown, a final set of filters and rules are applied:
    - **Impression Discounting:** Posts the member has seen recently are heavily down-ranked or removed (systems like Pinot track this).
    - **Content Moderation:** Any policy-violating content is removed.
    - **Frequency Caps/Anti-Gaming:** Rules to prevent a member from seeing too many posts from the same person, or too much of the same type of content in a short period.
- **So what?**: Freshness is critical. Repetitive content is actively penalized.
- **Now what?**:

- **Post Consistently, Not Repetitively:** Maintain a good posting cadence, but avoid reposting the exact same content frequently.
- **Vary Your Content:** If you post often, try to vary your topics or formats to avoid triggering frequency caps for individual followers.

VI. Output: Final Ranked Feed

Step 14: Presentation & Delivery

- **What happens**: The final, ordered list of posts is formatted for the member's specific device (web, iOS, Android – likely using "Render Models") and delivered to their screen.
- **So what?**: The content has made it!
- **Now what?**: Observe the engagement your content receives. This feeds back into the system (Step 15).

VII. Offline & Supporting Systems (The Continuous Learning Loop)

Step 15: Logging, Training, Evaluation, & Deployment

- **What happens**:
  - All interactions (impressions, clicks, comments, shares, dwell time) with the served feed are logged (via Kafka to HDFS/Pinot).
  - This data is used to create massive training datasets (using Spark/Beam).
  - New versions of all the AI models (L0, L1, L2, Setwise) are continuously trained on these datasets (using ProML/DARWIN, TensorFlow/PyTorch, Horovod).
  - Models are evaluated offline, and then promising ones are A/B tested online (using XLNT).
  - Successful models are deployed, and MOO weights are retuned.
- **So what?**: The system is constantly learning and adapting based on how members interact with content. Your content's performance directly influences future model training and, therefore, future content distribution.
- **Now what?**:
  - **Analyze Your Performance:** Use LinkedIn analytics to understand what content resonates with your audience (views, engagements, follower growth).

- **Iterate and Adapt:** Use these insights to refine your content strategy. What worked well? What didn't?
- **Be Patient & Consistent:** Building an engaged audience and consistently getting good distribution takes time as the system learns your value and your audience's preferences.

LinkedIn Profile Checklist for Marketers & Creators

Guiding Principle: Your LinkedIn profile is a rich data source for LinkedIn's AI. Every piece of information you provide helps the system understand who you are, what you know, what you're interested in, and who your professional community is. The clearer and more aligned this information is with your goals, the better the AI can match you with opportunities, connections, and relevant audiences for your content.

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1. Profile Photo & Background Photo

- **Why:** While not explicitly stated for *feed ranking* in the provided papers, profile photos are crucial for trust, recognition, and general platform engagement which *indirectly* influences how your content might be perceived or how connection requests are received. For direct profile views/search, a clear photo aids human recognition. Some systems (like those detecting AI-generated faces) analyze image characteristics.
- **What:** Use a clear, professional headshot. Ensure your background photo is relevant to your profession or brand.
- **How:**
    - **Profile Photo:**
        - High-resolution, well-lit, face clearly visible, professional attire.
        - Avoid busy backgrounds or distracting elements.
        - Ensure it's a real photo of you (systems are getting better at detecting AI-generated or fake images).
    - **Background Photo:**
        - Use a high-quality image (e.g., 1584 x 396 pixels).
        - Reflect your industry, company brand, a professional achievement, or a relevant personal brand statement.
        - Ensure any text on the background image is legible and not cut off on different devices.

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2. Headline

- **Why:** Your headline is a highly weighted text field used for search (both by humans and internal systems like Galene/Hosted Search), keyword extraction, and

generating your "member embedding" (your digital fingerprint). Clear, keyword-rich headlines help the system categorize you accurately.

- **What:** Craft a concise, keyword-rich headline that clearly states who you are, what you do, and the value you bring, particularly highlighting your top skills and areas of expertise.
- **How:**
    - **Keywords First:** Include 2-3 primary keywords relevant to your role, industry, and target audience (e.g., "Generative AI for Marketing Specialist," "Data-Driven Content Strategist").
    - **Value Proposition:** Briefly state the value you offer or the problems you solve (e.g., "Helping B2B SaaS companies scale with AI-powered marketing").
    - **Use the Full Character Limit (220 characters):** Maximize the information you provide.
    - **Mirror Target Audience Language:** Use terms your desired connections or audience would search for.
    - **Update Regularly:** If your focus or top skills shift, update your headline.

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3. About (Summary) Section

- **Why:** This is a significant block of text that feeds into natural language processing (NLP) models (like DeText, BERT-based models, or newer LLMs for feature generation) to understand your expertise, interests, and professional narrative. It heavily influences your member embedding and how you're matched for content visibility, connections, and job recommendations.
- **What:** Write a compelling, detailed summary that elaborates on your skills, experience, achievements, and professional goals, incorporating relevant keywords naturally.
- **How:**
    - **Storytelling with Keywords:** Weave in keywords related to your top skills, industry, and desired areas of influence. Don't just list them; use them in context. (e.g., "Leveraged *generative AI for marketing* to increase lead conversion by X%...").
    - **Detail Achievements:** Quantify your accomplishments where possible. This provides concrete data points.
    - **Express Your "Why":** Share your passion and what drives you professionally. This can add a unique dimension to your "embedding."

- **Include a Call to Action (Implicit or Explicit):** What do you want people to do after reading your summary (e.g., connect, view your work, discuss specific topics)?
- **Structure for Readability:** Use short paragraphs, bullet points, and white space. While AI reads it all, humans do too!
- **Mention Key "Entities":** If you've worked with notable brands or on significant projects (that you can publicly share), mentioning them can strengthen associations in the Economic Graph.

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4. Experience Section

- **Why:** Each role description is parsed for keywords, skills, company information, and timelines. This data is critical for:
  - **Entity Linking:** Connecting you to specific companies and industries in the Economic Graph (LIquid).
  - **Skill Inference:** AI infers skills from your job descriptions even if not explicitly listed in the skills section.
  - **Career Trajectory Modeling:** Understanding your career path, which can influence recommendations.
  - **Member Embedding:** Your work history is a strong component of your overall "digital fingerprint."
- **What:** Detail each role with specific responsibilities, achievements, and the skills utilized, using industry-standard language and keywords.
- **How:**
  - **Official Company Names:** Link to the correct official LinkedIn Company Page for each employer. This solidifies your position in the Economic Graph.
  - **Accurate Titles & Dates:** Use precise job titles and employment dates.
  - **Achievement-Oriented Descriptions:** For each role, use bullet points to describe your key responsibilities and, more importantly, your accomplishments. Quantify these with numbers where possible (e.g., "Managed a team of 5 and increased sales by 15%").
  - **Embed Keywords and Skills:** Naturally weave in the skills and keywords relevant to that specific role and industry. Think about what terms a recruiter or potential connection looking for someone with your experience in that role would search for.
  - **Use STAR Method Principles:** (Situation, Task, Action, Result) for describing accomplishments, which inherently provides rich contextual data for AI.

- **Add Rich Media (if relevant):** Link to projects, publications, or presentations related to that role if it adds professional context.

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5. Education Section

- **Why:** Connects you to educational institutions within the Economic Graph, identifies fields of study, and degrees, which are all features for matching and recommendation algorithms. Alumni networks are a strong signal for connection recommendations.
- **What:** Accurately list your educational background, linking to the official LinkedIn Pages for your institutions.
- **How:**
  - **Official School Names:** Link to the correct LinkedIn University Page.
  - **Accurate Degrees & Fields of Study:** Use standardized terms.
  - **Dates Attended:** Provide accurate dates.
  - **Activities and Societies (Optional but useful):** If relevant, mention significant activities or societies, as these can contain keywords and show areas of interest/soft skills.
  - **Description (Optional):** Briefly mention key projects or academic achievements if they highlight relevant skills or expertise.

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6. Skills Section (Endorsements & Skill Badges)

- **Why:** This is a primary source for explicit skill data. The system uses these listed skills (and their endorsements) for:
  - **Direct Matching:** For job recommendations, connection suggestions, and content filtering.
  - **Skill Graph Linkage:** Connecting your profile to the canonical skills within LinkedIn's Skills Graph.
  - **Member Embedding:** Your explicitly listed and endorsed skills heavily influence your "interest fingerprint."
  - **Skill Badges (Assessments):** Verified skills likely carry more weight in certain contexts.
- **What:** Curate a comprehensive list of your relevant hard and soft skills. Seek endorsements for your most important skills and consider taking LinkedIn Skill Assessments.

- **How:**
    - **List Up to 50 Relevant Skills:** Prioritize the skills most relevant to your current role and future aspirations.
    - **Pin Top 3 Skills:** Your most critical skills should be pinned so they are immediately visible.
    - **Use Standardized Skill Terms:** LinkedIn suggests skills as you type; try to use these recognized terms for better mapping to the Skills Graph.
    - **Seek Endorsements Strategically:** Ask connections who can genuinely vouch for your proficiency in specific skills to endorse you. Endorsements from people skilled in that area themselves may carry more weight.
    - **Take LinkedIn Skill Assessments:** Passing these assessments adds a "verified skill" badge to your profile, which can be a strong signal of proficiency to both AI and human viewers.
    - **Keep Skills Updated:** As you develop new skills or shift focus, update this section.

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7. Recommendations

- **Why:** Text from recommendations (both given and received) is processed by NLP models. It provides qualitative evidence of your skills and work ethic, and keywords within recommendations can be extracted. The identity of who recommends you (and who you recommend) also strengthens graph connections.
- **What:** Request and give thoughtful, specific recommendations that highlight key skills and achievements.
- **How:**
    - **Request Specific Recommendations:** When asking for a recommendation, guide the person on what aspects of your work or specific skills you'd like them to highlight.
    - **Give Detailed Recommendations:** When recommending others, be specific about their contributions, skills, and impact. Use relevant keywords. This not only helps them but also reflects positively on you as a thoughtful professional.
    - **Personalize Requests:** Don't use generic requests.

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8. Honors & Awards / Publications / Certifications / Languages / etc.

- **Why:** These sections provide additional structured data points and keywords that enrich your profile, contribute to your member embedding, and can be used as filters or signals in search and recommendation. Certifications, especially from recognized bodies (like IBM, Google in the example), are strong signals of specific expertise.
- **What:** Thoroughly complete all relevant sections with accurate and specific information.
- **How:**
    - **Be Comprehensive:** If you have relevant honors, awards, publications, patents, courses, projects, or certifications, list them.
    - **Use Official Names:** For publications or certifications, use the exact official names and link to issuing organizations where possible.
    - **Keywords within Descriptions:** If there's a description field (e.g., for projects), use it to provide context and include relevant keywords.
    - **Language Proficiency:** Accurately list your language proficiencies. This is important for global networking and opportunities.

By meticulously optimizing these profile sections, marketers and creators provide LinkedIn's AI systems with clear, rich, and structured data. This, in turn, increases the likelihood of their profile being surfaced in relevant searches, their content reaching the right audience, and them being suggested as valuable connections or candidates.

LinkedIn Content Pre-Launch Checklist for Creators

Guiding Principle: Every piece of content you post is an input to LinkedIn's sophisticated AI. The system analyzes its substance, format, and the engagement it generates to determine its relevance and value to other members. Your goal is to create content that is easily understood by the AI, highly engaging for your target audience, and signals value within the LinkedIn ecosystem.

I. Before You Post: Content Strategy & Creation

1. Topic Selection & Keyword Research

- **Why:** Content is analyzed for topics and keywords during "Inputs Gathering & Preprocessing" (Step 1). This information heavily influences its initial "digital fingerprint" (embedding) and how it's categorized. This impacts its chances in "Similarity-Based Candidate Generation" (Step 5) and relevance scoring in "L2 Rich Ranking" (Step 8 & 9).
- **What:** Strategically choose topics and keywords that align with your target audience's interests and your expertise, and for which your content can provide unique value.
- **How:**
    - **Audience Pain Points:** Identify the key questions, challenges, and interests of your target professional audience. What information are they actively seeking on LinkedIn?
    - **Keyword Research (LinkedIn Context):**
        - Use LinkedIn's search bar to see what terms people are using related to your topic. Observe the "Searches related to..." suggestions.
        - Analyze popular posts (yours and others') in your niche. What keywords and hashtags are common?
        - Consider what terms a member of your target audience would type into search to find information your content provides.
    -

- **Content Gaps:** Look for underserved topics or unique angles on existing topics where you can offer fresh insights. This increases your chances of standing out in "Setwise Re-Ranking" (Step 10).
- **Align with Your Profile:** Ensure content topics are consistent with the expertise and skills showcased on your profile (influences author credibility signals in L2).

2. Content Format Selection

- **Why:** Different content formats (text, image, video, article, document/carousel, poll) are processed and potentially weighted differently by the ranking algorithms. Some formats inherently encourage different types of engagement (e.g., videos for dwell time, polls for quick interaction). The system learns which formats your audience prefers.
- **What:** Choose a content format that best suits your message and is known to engage your target audience. Experiment with different formats.
- **How:**
  - **Text Posts:** Ideal for quick insights, questions, or starting discussions. Ensure they are well-structured and easy to read.
  - **Images:** Use high-quality, relevant images that enhance your message. Provide descriptive alt-text (accessibility and keyword signal). The system analyzes image content.
  - **Videos (Native):** LinkedIn often prioritizes native video. Keep them concise and engaging, especially in the first few seconds. Add captions (accessibility and helps AI understand content). Video content is analyzed. Aim for formats that encourage "Long Dwell" (Step 9).
  - **Documents/Carousels (PDFs):** Excellent for sharing multi-page insights, presentations, or guides. Make them visually appealing and easy to scan. Each page can be a new opportunity for keywords.
  - **Articles/Newsletters (LinkedIn Publishing Platform):** Best for long-form, in-depth content. Well-written articles can establish thought leadership. They have their own distribution potential.
  - **Polls:** Great for quick engagement and gathering opinions. Signals active interaction (Step 9).
  - **Analyze Competitors/Influencers:** What formats are successful for similar creators targeting your audience?

    - **Test and Iterate:** Monitor your analytics to see which formats perform best with *your* specific audience.

3. Crafting High-Quality, Engaging Content

- **Why:** The "L2 Rich Ranking" (Step 9) predicts multiple engagement probabilities (P(Like), P(Comment), P(Share), P(Long Dwell), P(Skip)). High-quality, genuinely engaging content is more likely to trigger these positive signals and avoid negative ones (like P(Skip)). "Dwell time" is a key implicit signal.
- **What:** Create content that is valuable, insightful, well-structured, and encourages meaningful interaction from your target audience.
- **How:**
    - **Value First:** Provide genuine value – insights, solutions, new perspectives, helpful resources.
    - **Originality & Authenticity:** Share your unique perspective. Authentic content tends to resonate more.
    - **Clarity & Conciseness:** Get to the point. Even for long-form content, ensure it's well-organized. Use clear language.
    - **Hook Attention Early:** The first few lines of a text post or seconds of a video are critical to avoid a "skip."
    - **Encourage Discussion:** Ask open-ended questions. Invite comments and different viewpoints. This directly influences P(Comment).
    - **Visual Appeal:** For posts with visuals, ensure they are high quality and complementary. For text-heavy posts (like articles or documents), use formatting (bolding, bullets, subheadings) to improve readability and dwell time.
    - **Proofread:** Typos and grammatical errors can reduce credibility and engagement.
    - **Mobile-First Design:** Many members access LinkedIn on mobile. Ensure your content is easily consumable on smaller screens (e.g., legible font sizes in images/documents, concise video intros).

II. As You Post: Optimizing for Discovery & Initial Engagement

4. Writing Compelling Copy & Headlines

- **Why:** The text of your post (including headlines for articles/videos) is a primary input for NLP analysis and embedding generation (Step 1). Clear, engaging copy with relevant keywords improves initial understanding by the AI and attracts human attention, impacting P(Click) and P(Long Dwell).
- **What:** Craft clear, concise, and compelling text that includes relevant keywords and encourages viewers to engage further.
- **How:**
    - **Strong Opening:** Make the first sentence or two captivating to stop the scroll.
    - **Incorporate Keywords Naturally:** Weave in 1-3 primary keywords your audience would search for or associate with the topic.
    - **Clear Call to Action (CTA) (if applicable):** What do you want people to do? (e.g., "What are your thoughts?", "Read the full article here:", "Share your experience below.")
    - **Use Emojis Sparingly & Strategically:** They can add visual interest but overuse can look unprofessional.
    - **For Articles/Newsletters:** Write a strong, keyword-rich headline and an engaging introductory paragraph.

5. Strategic Use of Hashtags

- **Why:** Hashtags are explicit categorization signals used in "Inputs Gathering" (Step 1) and contribute to your post's "digital fingerprint." They help connect your content to broader topic discussions and interest-based communities, influencing "Graph-Based" and "Similarity-Based Candidate Generation" (Steps 4 & 5). Your "Viewer Hashtag IDs" and "Actor Hashtag IDs" are even specific features (mentioned in LiRank).
- **What:** Use a mix of broad and niche hashtags relevant to your content and target audience.
- **How:**
    - **3-5 Relevant Hashtags:** This is generally a good range. Too many can look spammy.
    - **Mix Broad and Niche:**

        - *Broad:* e.g., #marketing, #leadership (helps with wider discovery).
        - *Niche:* e.g., #GenerativeAIMarketing, #B2BSaaSContent (helps reach a more specific, interested audience).
    - **Research Hashtags:** See which hashtags are popular and relevant in your industry. Check the follower counts for hashtags.
    - **Branded Hashtag (if applicable):** If you have one, use it consistently.
    - **Avoid Irrelevant Hashtags:** This can harm your content's perceived relevance.

6. Tagging Relevant People & Companies (When Appropriate)

- **Why:** Tagging explicitly links your content to other entities in the Economic Graph (Step 3), potentially increasing its reach to their networks and strengthening signals for "Graph-Based Candidate Generation" (Step 4). It can also trigger notifications, encouraging initial engagement.
- **What:** Tag individuals or companies only when they are genuinely relevant to the content (e.g., you're referencing their work, they are collaborators, or the content is directly about them).
- **How:**
    - **Relevance is Key:** Don't tag people or companies just for visibility if they have no connection to the post. This can be perceived as spammy by both users and the algorithm.
    - **Notify & Engage:** If you tag someone, they'll be notified. This can spark initial engagement if they find the content relevant.
    - **Company Mentions:** Tagging your own company or relevant partner companies can increase visibility within those company networks.

7. Choosing the Right Posting Time

- **Why:** While the system aims for relevance over pure chronology, posting when your target audience is most active on LinkedIn can increase the chances of initial engagement. These early engagement signals (likes, comments, shares, dwell time) are critical features for the "L2 Rich Ranking" (Step 9) and can influence wider distribution.
- **What:** Experiment with posting at different times and days to find when your specific audience is most active and responsive.

- **How:**
    - **General Guidelines (Starting Point):** Weekday mornings and lunchtimes are often cited as good, but this varies greatly by industry and audience.
    - **Use LinkedIn Analytics:** Check your post analytics to see when your existing audience engages most.
    - **Consider Time Zones:** If you have a global audience, you might need to vary posting times or post multiple times.
    - **Test and Observe:** Post at different times over a few weeks and track initial engagement velocity.

III. After You Post: Fostering Engagement & Learning

8. Engaging with Comments Promptly & Thoughtfully

- **Why:** Comments are a strong positive engagement signal for "L2 Rich Ranking" (P(Comment) in Step 9). Your replies keep the conversation going, increasing the overall engagement on the post, which can lead to further distribution. It also signals to the algorithm that the content is fostering a valuable discussion.
- **What:** Monitor your posts for comments and respond to them in a timely and thoughtful manner to encourage further discussion.
- **How:**
    - **Acknowledge Comments:** Even a simple "thanks for sharing" can be valuable.
    - **Answer Questions:** If people ask questions, provide helpful answers.
    - **Foster Debate (Respectfully):** If there are differing opinions, facilitate a respectful discussion.
    - **Tag Commenters in Replies (if appropriate):** This ensures they see your response.
    - **Don't Be Afraid to Ask Follow-up Questions:** Keep the conversation flowing.

9. Analyzing Post Performance

- **Why:** This is part of the "Logging, Training, Evaluation, & Deployment" loop (Step 15). Your content's performance (views, likes, comments, shares, demographics of viewers) provides data that trains future iterations of LinkedIn's AI models. Understanding what works helps you create better content that the system will favor.
- **What:** Regularly review your LinkedIn content analytics to understand what topics, formats, and approaches resonate most with your audience and lead to the best distribution.
- **How:**
    - **Track Key Metrics:** Views, click-through rates (for links), comment count, like count, reshare count, engagement rate.
    - **Identify Top Performing Content:** What did your most successful posts have in common (topic, format, headline, hashtags, posting time)?
    - **Identify Underperforming Content:** What didn't work? Try to understand why.
    - **Look at Viewer Demographics:** Are you reaching your intended audience?
    - **Adjust Strategy Based on Data:** Use these insights to refine your content strategy for future posts.

10. Resharing Your Valuable Evergreen Content (Strategically)

- **Why:** Not everyone in your network will see your post the first time due to algorithmic filtering and timing. Resharing (with added context or updates) gives valuable evergreen content another chance to be picked up by the "Candidate Generation" stages (Steps 4-6) and be seen by a different segment of your audience or new followers.
- **What:** Periodically reshare your most valuable and timeless content, perhaps with a new insight or update, to extend its reach.
- **How:**
    - **Don't Just Hit "Reshare":** Add new commentary, an update, or ask a new question when resharing to provide fresh value.
    - **Space It Out:** Don't reshare content too frequently. Wait weeks or months.

- **Consider Different Times/Days:** Reshare at a different time than the original post to potentially reach a different segment of your audience.
- **Track Performance:** See if the reshared content gains new traction.

By following this comprehensive checklist, marketers and creators can significantly improve how their content interacts with LinkedIn's sophisticated feed algorithms, leading to better visibility, more meaningful engagement, and ultimately, greater impact.

LinkedIn Engagement Checklist for Marketers and Creators

Guiding Principle: Your engagement activities on LinkedIn are critical data points. They signal your interests, your expertise, your network's relevance, and your value as a community member to LinkedIn's AI. Thoughtful, consistent, and authentic engagement helps refine your "member embedding," influences how others perceive you (and thus engage with your content), and can directly impact the visibility of your own contributions through various system heuristics and graph signals.

I. Quick Daily Engagements (5-15 minutes per day)

1. Reacting Strategically to Relevant Feed Content

- **Why:** Your reactions (Like, Celebrate, Support, Love, Insightful, Funny) are activity signals fed into "Inputs Gathering" (Step 2) and influence your member embedding. They also provide feedback to the "L2 Rich Ranking" (Step 9) of the content you react to. Reacting to content from your target audience or in your niche strengthens your association with those topics and individuals in the Economic Graph (Step 3) and for "Similarity-Based Candidate Generation" (Step 5).
- **What:** Quickly scan your feed and thoughtfully react to 3-5 posts that are highly relevant to your expertise, industry, or target audience.
- **How:**
    - **Prioritize Relevance:** Focus on posts from key connections, industry leaders, or on topics central to your brand.
    - **Use Diverse Reactions:** Don't just "Like" everything. Use "Insightful," "Celebrate," or "Support" where appropriate to convey more nuanced engagement. This richer signal might be captured by the system.
    - **Timeliness:** Reacting to newer, relevant posts can be more impactful.
    - **Avoid indiscriminate reacting:** Mass liking/reacting can dilute the signal of your true interests.

2. Skimming and Acknowledging Relevant Notifications

- **Why:** Notifications (e.g., someone commented on your post, endorsed your skill, viewed your profile, work anniversaries) are direct prompts for re-engagement. Clicking through or interacting with them signals active platform use, which is a

positive behavioral feature. Some "Heuristic-Based Candidate Generation" (Step 6) might prioritize content from members who are currently active or responsive.

- **What:** Briefly review your notifications and acknowledge or take quick action on the most relevant ones.
- **How:**
    - **"Thanks for the endorsement" or "Congrats on the new role!":** Quick, personalized replies to relevant notifications.
    - **Acknowledge Profile Views (if appropriate for your strategy):** This can sometimes initiate a connection.
    - **Respond to mentions:** If someone tags you, acknowledge it.
    - **Quickly clear irrelevant notifications:** Shows you're actively managing your presence.

3. Brief, Insightful Comments on 1-2 Key Posts

- **Why:** Comments are a very strong engagement signal, heavily weighted in "L2 Rich Ranking" (P(Comment) in Step 9) and the "MOO Combination" (Step 12). Leaving relevant comments on others' posts establishes your expertise, increases your visibility to their network ("Graph-Based Candidate Generation" Step 4 for your *own* future content), and can lead to new connections. Your comment text is also analyzed (NLP, embeddings).
- **What:** Identify 1-2 highly relevant posts in your feed (or from key influencers/companies you follow) and add a brief, thoughtful comment that contributes to the discussion.
- **How:**
    - **Add Value, Don't Just Agree:** Expand on a point, ask a clarifying question, or share a related brief experience.
    - **Use Relevant Keywords (Naturally):** Your comment text becomes associated with your profile and the topic.
    - **Tag the Author (Optional):** @mention the author if you're directly responding to their point, to ensure they see it.
    - **Keep it Professional and Constructive:** Align with LinkedIn's community policies.
    - **Be Timely:** Commenting on fresher posts often yields more visibility.

II. Focused Daily/Regular Engagements (15-30 minutes per day/several times a week)

4. Participating Actively in 1-2 Relevant LinkedIn Groups

- **Why:** Group activity signals strong interest in specific niches. Content shared within groups, and your interactions there, contribute to your member embedding and can influence "Similarity-Based Candidate Generation" (Step 5) for both content and connections. Posts from groups you are active in might also get prioritized by some heuristics.
- **What:** Identify and actively participate in 1-2 LinkedIn Groups that are highly relevant to your industry, expertise, or target audience.
- **How:**
    - **Share Valuable Content (Your Own or Curated):** Post relevant articles, insights, or questions within the group.
    - **Engage with Others' Posts:** Like, comment, and answer questions in group discussions.
    - **Contribute Expertise:** Position yourself as a helpful resource within the group.
    - **Choose Active, Well-Moderated Groups:** Quality of interaction matters more than quantity of groups.
    - **Follow Group Rules:** Ensure your contributions are aligned with the group's purpose and guidelines.

5. Sending Personalized Connection Requests

- **Why:** Expanding your relevant network strengthens your position in the Economic Graph (Step 3), which directly impacts "Graph-Based Candidate Generation" (Step 4) for your content and profile visibility. Personalized requests have a higher acceptance rate, and accepted connections are strong positive signals.
- **What:** Send a few targeted, personalized connection requests each day/week to individuals relevant to your professional goals.
- **How:**
    - **Always Add a Personal Note:** Explain *why* you want to connect (e.g., "I enjoyed your recent post on X," "We share an interest in Y," "I see we both worked at Z company").
    - **Focus on Mutual Value:** Think about what value the connection might bring to *them* as well.
    - **Quality over Quantity:** A few thoughtful connection requests are better than many generic ones.

    - **Connect with People Who Engage with Your Content:** If someone consistently likes or comments on your posts, they might be a good connection.
    - **Leverage "People You May Know":** This feature is already using AI to suggest relevant connections.

6. Engaging in LinkedIn Messaging (Direct Messages & Group Chats)

- **Why:** While private, messaging activity (frequency, responsiveness) can be an indicator of overall platform engagement, which might be a feature in broader activity models. Building relationships via messaging can lead to more public engagement (e.g., someone you chat with is more likely to engage with your feed posts). Group chats signal community involvement.
- **What:** Use LinkedIn Messaging to nurture professional relationships, follow up on connections, and participate in relevant group chats.
- **How:**
    - **Follow Up After Connecting:** Send a brief message after a new connection is made.
    - **Share Relevant Information Privately:** If you find an article or resource that a specific connection might find valuable, share it via message.
    - **Be Responsive:** Try to respond to messages in a timely manner.
    - **Participate in Relevant Group Chats:** If you're invited to or create group chats around professional topics, contribute meaningfully.
    - **Maintain Professionalism:** All communication should be professional.

III. More Involved Weekly/Bi-Weekly Engagements (30-60+ minutes per session)

7. Writing and Publishing LinkedIn Articles or Newsletters

- **Why:** Long-form content like articles and newsletters allows for deep exploration of topics, generating rich text for NLP analysis and strong "post embeddings" (Step 1 & 5). Successful articles/newsletters can attract followers to your profile/newsletter, directly growing your audience and strengthening your graph signals (Step 3 & 4). They signal expertise and thought leadership.
- **What:** If you have in-depth insights to share, consider publishing LinkedIn Articles or starting a Newsletter on a topic relevant to your expertise and target audience.
- **How:**

    - **Choose a Niche Focus:** Consistent topics help build a dedicated audience.
    - **Provide Substantial Value:** Articles/newsletters should offer deep insights, comprehensive guides, or unique perspectives.
    - **Optimize for Readability:** Use headings, subheadings, bullet points, and images.
    - **SEO Principles Apply:** Use relevant keywords in your title and throughout the text.
    - **Promote Your Article/Newsletter:** Share it as a post, in relevant groups, and potentially in your email signature.
    - **Engage with Comments:** Foster discussion on your published pieces.
    - **Consistency is Key:** If you start a newsletter, try to maintain a regular publishing schedule.

8. Participating in or Hosting LinkedIn Live/Audio Events

- **Why:** Live events generate significant engagement signals in real-time (views, comments, reactions) and can have high dwell times (P(Long Dwell) in Step 9). They position you as an active expert and content shared (even transcripts/recordings) can be further analyzed. Successful events can attract new followers.
- **What:** Consider participating as a speaker in relevant LinkedIn Live/Audio events, or host your own if you have a strong topic and audience.
- **How:**
    - **Preparation is Crucial:** Have a clear agenda, engaging content, and practice your delivery.
    - **Promote Extensively Beforehand:** Let your network know about the event.
    - **Engage with the Audience During the Event:** Acknowledge comments, answer questions live.
    - **Repurpose Content:** Share the recording or key takeaways as a follow-up post.
    - **Technical Setup:** Ensure good audio/video quality.
    - **Start as a Guest:** If hosting seems daunting, offer to be a guest on an established Live/Audio event in your niche to gain experience.

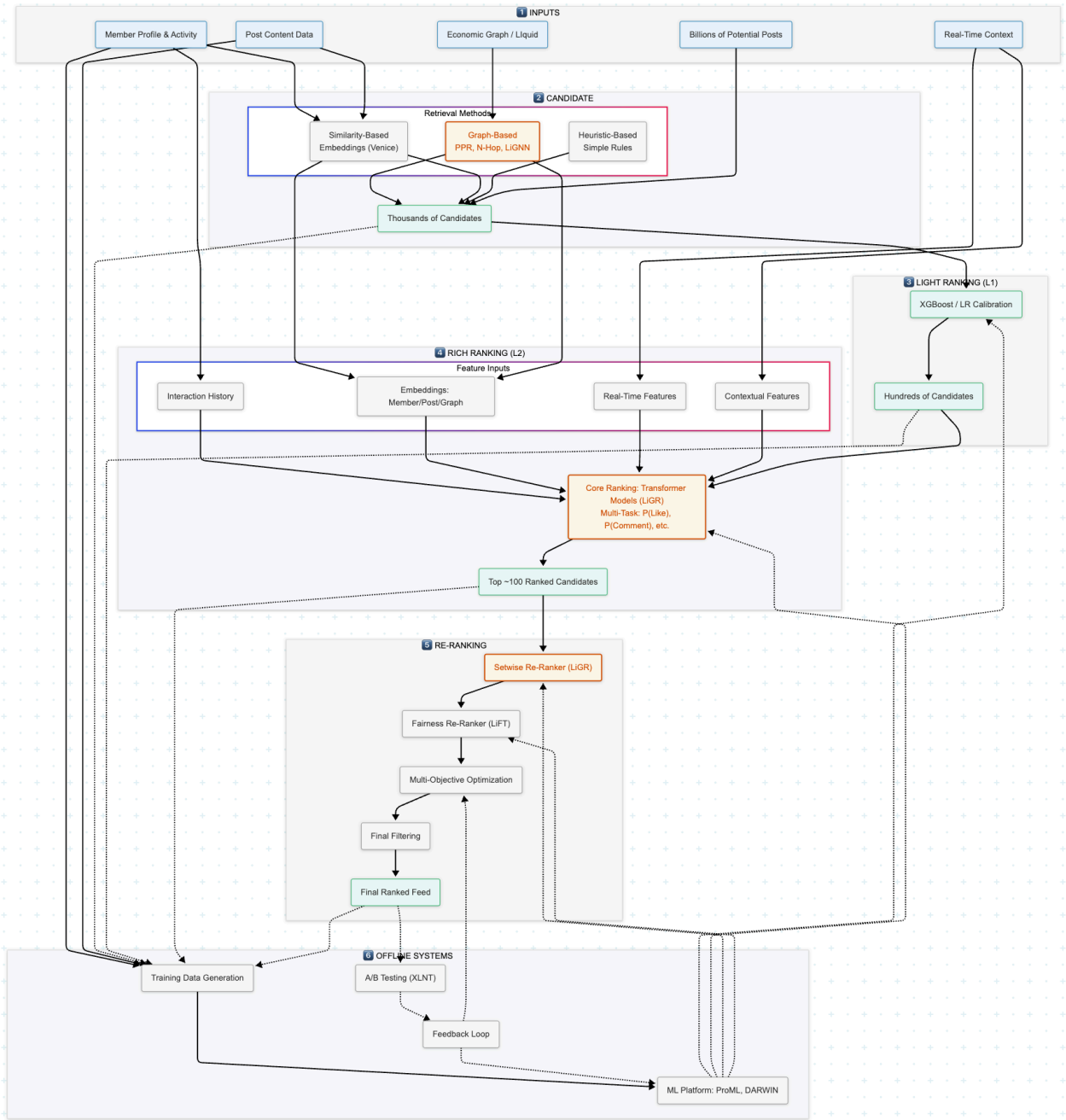
9. Reviewing and Endorsing Skills for Connections

- **Why:** Endorsing skills strengthens the Skill Graph component of the Economic Graph (Step 3). While primarily benefiting the person you endorse, this reciprocal activity can also signal your own areas of expertise and your engagement within your professional community. It's a form of positive network interaction.

- **What:** Periodically review connection profiles and endorse skills for which you can genuinely vouch.
- **How:**
    - **Be Authentic:** Only endorse skills you know the person possesses.
    - **Focus on Key Skills:** Prioritize endorsing the most relevant skills for your connections.
    - **Reciprocity Often Occurs:** Connections you endorse may be more likely to endorse you back.

By consistently applying these engagement strategies, creators and marketers actively provide positive signals to LinkedIn's algorithms across multiple stages. This isn't just about "being active" – it's about strategic, relevant, and valuable interactions that align with how the platform technologically identifies and distributes professional content and fosters community.

LinkedIn Newsfeed System Diagram



LinkedIn Newsfeed Technologies

Here's an outline of the LinkedIn newsfeed algorithm based on the data we extracted from LinkedIn. If you've wondered, as you read this paper, where we got the information from, or how we know certain things, it's all from LinkedIn's engineering documents.

I. Inputs Gathering & Preprocessing

1. **A. Billions of Potential Posts:**
    - 1.1. **Data Ingestion:** Raw post data (text, images, videos, links, metadata like creation time, author ID) is ingested from various creation points into primary storage systems (e.g., Espresso, object storage like Ambry).
    - 1.2. **Content Understanding & Enrichment (Nearline/Batch):**
        - 1.2.1. **Text Processing (DeText/LLMs - e.g., EON/360Brew for feature generation):**
            - 1.2.1.1. Tokenization, normalization, language detection.
            - 1.2.1.2. Entity extraction (skills, companies, topics).
            - 1.2.1.3. Sentiment analysis.
            - 1.2.1.4. Generation of dense text embeddings (e.g., using multilingual-e5, LiBERT, or newer foundation models).
        - 1.2.2. **Image/Video Processing:**
            - 1.2.2.1. Feature extraction (object detection, scene understanding).
            - 1.2.2.2. Generation of multimedia embeddings.
        - 1.2.3. **Metadata Standardization:** Standardizing post types, categories.
    - 1.3. **Feature Storage (Venice/Feature Cloud):** Processed post features and embeddings are stored in online key-value stores for fast retrieval.
2. **B. Member Profile & Activity History:**
    - 2.1. **Profile Data (Espresso/Liquid):**
        - 2.1.1. Static attributes: Industry, location, skills, education, title, company, connections.
        - 2.1.2. Standardized entities derived from profile (e.g., standardized skills from Skills Graph).
    - 2.2. **Activity Data (Kafka -> Pinot/Venice/HDFS):**
        - 2.2.1. **Event Ingestion (Kafka):** Raw interaction events (views, clicks, likes, comments, shares, searches, job applies, profile views, dwell time on items) are streamed.

        - 2.2.2. **Near Real-Time Processing (Samza/Beam on Samza -> Pinot/Venice):**
            - 2.2.2.1. Short-term activity aggregation (e.g., actions in last N hours/days).
            - 2.2.2.2. Sessionization.
            - 2.2.2.3. Storage of recent activity features for fast access.
        - 2.2.3. **Batch Processing (Spark/Beam on Spark from HDFS -> Venice/Feature Cloud):**
            - 2.2.3.1. Long-term activity aggregation and feature engineering (e.g., affinities, historical engagement patterns).
            - 2.2.3.2. Generation of member embeddings based on historical activity.
    - 2.3. **Pensieve/Feature Cloud:** Serves pre-computed member embeddings.
3. **C. Economic Graph / LIquid:**
    - 3.1. **Data Ingestion:** Updates to graph structure (connections, follows, company affiliations, skill relations) are ingested into LIquid.
    - 3.2. **Graph Maintenance:** LIquid maintains the graph representation, enabling traversals and queries.
4. **D. Post Content Data:** (Covered by A.1)
5. **E. Real-Time Context:**
    - 5.1. **Session Information:** Device type, time of day, current location (if available).
    - 5.2. **Current Query/Intent:** If feed is being triggered by a specific search or navigation.

II. L0: Candidate Generation (Recall-Focused)

1. **A. Graph-Based Retrieval:**
    - 1.1. **LIquid Queries:**
        - 1.1.1. **N-Hop Neighbor Retrieval:** Execute graph traversal queries on LIquid to find 2nd/3rd degree connections, etc., for the viewing member.
        - 1.1.2. **Personalized PageRank (PPR) Calculation:** Execute PPR algorithms (potentially pre-computed or nearline updated) within or using LIquid to find influential nodes in the member's vicinity.
    - 1.2. **GNN Embedding Based Retrieval (LiGNN, Feature Cloud/Venice, Hosted Search/EBR System):**
        - 1.2.1. **Member GNN Embedding Fetch (Venice/Pensieve):** Retrieve pre-computed GNN embedding for the viewing member.

        - 1.2.2. **Approximate Nearest Neighbor (ANN) Search (Hosted Search/FAISS/ScaNN on item GNN embeddings):** Use the member GNN embedding to query an index of GNN embeddings for all posts/authors.
        - 1.2.3. Retrieve top-K candidates based on embedding similarity.
2. **B. Similarity-Based Retrieval (EBR):**
    - 2.1. **Member Embedding Fetch (Venice/Pensieve/Feature Cloud):** Retrieve pre-computed multi-modal member embedding (based on profile, activity, etc.).
    - 2.2. **Item Embedding Index (Hosted Search/Zelda/Custom ANN):** Access an index of pre-computed item (post) embeddings stored efficiently (e.g., in Venice, served via an EBR system like Zelda).
    - 2.3. **ANN Search:** Perform ANN search using the member embedding against the item embedding index.
    - 2.4. Retrieve top-K candidates based on embedding similarity (e.g., cosine similarity, dot product).
3. **C. Heuristic-Based Retrieval:**
    - 3.1. **Rule Engine/Simple Queries (Pinot/Venice):**
        - 3.1.1. Query for posts from recently interacted-with authors.
        - 3.1.2. Query for recently created popular content in member's network/groups.
        - 3.1.3. Query for content matching specific contextual triggers.
4. **D. Candidate Aggregation & Deduplication:**
    - 4.1. Collect candidates from all L0 sources.
    - 4.2. Deduplicate candidates appearing from multiple sources.
    - 4.3. Output: Thousands of unique candidate post IDs.

III. L1: Light Ranking (Calibration & Recall)

1. **A. Feature Fetching (Venice/Pinot):**
    - 1.1. For each L0 candidate post, fetch lightweight features (e.g., basic post metadata, author metadata, raw engagement counts, L0 source scores).
    - 1.2. Fetch lightweight member features.
2. **B. Scoring (Lightweight Model - e.g., XGBoost, Logistic Regression, shallow DNN executed via Quasar/ProML):**
    - 2.1. Input features into the L1 ranking model.
    - 2.2. Model predicts a preliminary relevance score (e.g., P(Click) or a combined engagement proxy).
    - 2.3. Calibrate scores from different L0 sources if necessary.

3. **C. Candidate Selection:**
    - 3.1. Rank candidates by L1 scores.
    - 3.2. Select top-K (hundreds) candidates.
    - 3.3. Output: Hundreds of candidate post IDs + L1 scores.

IV. L2: Rich Ranking / SPR (Precision-Focused) - Core Selection Logic

1. **A. Feature Engineering & Retrieval (Venice, Pinot, Feature Cloud, Frame):**
    - 1.1. **Member History Sequence Construction:**
        - 1.1.1. Retrieve recent member interaction sequence (post IDs, action types, timestamps, associated entity IDs like author/hashtag) from near real-time stores (Pinot/Venice).
        - 1.1.2. Tokenize/embed these historical items and actions.
    - 1.2. **Candidate Post Feature Fetching (Venice):** For each L1 candidate, retrieve rich features:
        - 1.2.1. Dense content embeddings (text, image, video).
        - 1.2.2. Detailed metadata, standardized entities.
        - 1.2.3. Author features and embeddings.
    - 1.3. **Member Feature Fetching (Venice/Pensieve):**
        - 1.3.1. Rich member profile embeddings.
        - 1.3.2. Long-term aggregated activity features.
        - 1.3.3. GNN-derived member embeddings.
    - 1.4. **Interaction/Pair Features (Computed on-the-fly or fetched from Venice):**
        - 1.4.1. Member-candidate author affinity.
        - 1.4.2. Member-candidate topic/skill similarity using embeddings.
    - 1.5. **Real-Time Contextual Features (Pinot/Session Data):** Current session intent, device, location.
2. **B. Core Ranking Model Inference (Transformer Sequential Models - LiGR, executed on ProML/Model Cloud, potentially leveraging TensorFlow/PyTorch):**
    - 2.1. **Input Preparation:** Feed member history sequence, candidate post features, member features, and contextual features into the model.
    - 2.2. **Transformer Encoding:** Process member history sequence through Transformer layers to get contextualized member representation.
    - 2.3. **Interaction Layers:** Interact member representation with candidate post representations (e.g., via attention, concatenation followed by MLPs).
    - 2.4. **Prediction Heads (Multi-Task Learning):**
        - 2.4.1. Predict P(Like) for the candidate.
        - 2.4.2. Predict P(Comment).

        - 2.4.3. Predict P(Share).
        - 2.4.4. Predict P(Click).
        - 2.4.5. Predict P(Long Dwell) / Dwell Time.
        - 2.4.6. Predict P(Skip).
        - 2.4.7. Other engagement/value predictions.
    - 2.5. **Isotonic Calibration Layer (Applied within DNN):** Calibrate raw model outputs.
    - 2.6. Output: Top ~100 ranked candidates with a vector of predicted probabilities for each action.

V. Re-Ranking & Finalization

1. **A. Setwise Re-Ranking (LiGR model component, executed on ProML/Model Cloud):**
    - 1.1. **Input:** Top-K (e.g., 10-20) candidates and their L2 scores/features from the pointwise model.
    - 1.2. **Transformer-based Setwise Attention:** Model interactions *between* candidates in the slate.
    - 1.3. **Score Adjustment:** Modify pointwise scores based on slate context to improve diversity, reduce redundancy, and enhance overall session coherence.
    - 1.4. Output: Adjusted scores for the top-K candidates.
2. **B. Fairness Re-Ranking (LiFT-based components/ProML):**
    - 2.1. Input: Candidates and their (potentially setwise-adjusted) scores.
    - 2.2. Retrieve member protected attributes (privacy-preserving manner, PPRE).
    - 2.3. Apply fairness algorithms (e.g., re-ranking, score adjustments) to mitigate identified biases and ensure equitable opportunity distribution.
    - 2.4. Output: Fairness-adjusted scores/ranking.
3. **C. Multi-Objective Optimization (MOO) Combination:**
    - 3.1. **Input:** Vectors of predicted probabilities (P(Like), P(Click), etc.) for each candidate, potentially after fairness adjustments.
    - 3.2. **Weighted Summation:** Apply pre-defined or dynamically tuned (Bayesian Optimization) weights to each predicted probability: Final_Score = w1*P(Like) + w2*P(Comment) + ... - w_skip*P(Skip) + ...
    - 3.3. Output: A single composite score for each candidate.
4. **D. Final Filtering & Business Rules:**
    - 4.1. **Impression Discounting (Pinot for recent impressions):** Down-rank or filter out content the member has recently seen.

    - 4.2. **Block Lists/Content Moderation Filters:** Remove policy-violating or low-quality content that may have slipped through.
    - 4.3. **Frequency Capping/Anti-Gaming Rules:** Apply rules to prevent over-exposure from a single source or topic.
    - 4.4. Output: Final ordered list of posts.

VI. Output: Final Ranked Feed

1. **A. Presentation Layer Formatting:** Prepare the selected posts for display on the specific client (web, iOS, Android) using Render Models.
2. **B. Delivery to Client:** Send the formatted feed data to the member's device.

VII. Offline & Supporting Systems (Continuous Loop)

1. **A. Logging & Data Collection (Kafka -> HDFS/Pinot):**
    - 1.1. Log impressions, clicks, dwells, all interactions on the served feed.
    - 1.2. Log features used at scoring time.
    - 1.3. Log A/B experiment assignments.
2. **B. Training Data Generation (Spark/Beam on Spark on HDFS):**
    - 2.1. Join interaction logs with feature snapshots.
    - 2.2. Generate (member, item, context, label) tuples for each objective.
    - 2.3. Apply sampling strategies (e.g., inverse propensity scoring, negative sampling).
    - 2.4. Prepare data in TFRecord or other training-efficient formats.
3. **C. Model Training (ML Platform - ProML/DARWIN, leveraging TensorFlow/PyTorch, Horovod for distributed training):**
    - 3.1. Train L0 embedding models (Two-Tower, GNNs).
    - 3.2. Train L1 light rankers.
    - 3.3. Train L2 core ranking models (Transformer-based MTL).
    - 3.4. Train Setwise re-rankers.
    - 3.5. Incremental training for freshness where applicable.
4. **D. Model Evaluation (Offline Metrics & Human Review):**
    - 4.1. Calculate AUC, NCE, Recall@k, Precision@k, fairness metrics on holdout sets.
    - 4.2. Conduct human relevance and quality reviews.
5. **E. Model Deployment (ML Platform to Model Cloud/Venice/Online Systems):**
    - 5.1. Version and push trained model artifacts (graphs, embeddings, weights) to serving systems.
6. **F. A/B Testing (XLNT Platform):**
    - 6.1. Deploy new models/features to a subset of members.

    - 6.2. Collect online engagement metrics (DAU, sessions, CTR, contribution rates, dwell time, fairness metrics).
    - 6.3. Statistical analysis to determine impact.
7. **G. MOO Parameter Tuning (Offline Simulation & Online Bayesian Optimization):**
    - 7.1. Adjust weights in the MOO function based on A/B test results and strategic objectives.

This detailed breakdown illustrates the complexity and interplay of various technologies and algorithmic stages involved in delivering a personalized and relevant LinkedIn Feed.

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Methodology and Disclosures

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We used Google's Gemini 2.5 Advanced 0504 model to synthesize this guide from the data. The source data is approximately 275,000 words.